CompTIA N+ · Network Security, Maintenance & Troubleshooting Procedures

M2-A1 Full Assessment — Nexus Innovations

SECTION A — Concept Application

Answer all questions. Each question presents a real-world scenario followed by a concept-specific question.

S1 — Selecting CPU, RAM and Storage for Multi-Tasking Workstations

Scenario: Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time.

Answer:

CPU: Choose a modern multi-core processor (at least 6–8 cores / 12–16 threads) with high single-thread performance and good multi-thread capability (e.g. Intel Core i5/i7 or AMD Ryzen 5/7). This allows the IDE, browser tabs and video-conferencing app to run concurrently without lag. Check motherboard socket compatibility (LGA 1700, AM5, etc.).

RAM: Minimum 16 GB DDR4/DDR5; 32 GB preferred for heavy multitasking. Ensure the modules match the motherboard's supported speed, voltage and dual-channel configuration. Matched pairs avoid timing errors.

Storage: NVMe SSD (512 GB or larger) for the OS and applications to give fast boot and load times. Optionally add a second SATA SSD or HDD for project files. Confirm the motherboard has the required M.2 or SATA connectors and that the drive is on the supported list.

Compatibility checklist: motherboard chipset & socket, RAM speed/capacity limits, PSU wattage sufficient for the chosen CPU + GPU (if any), case clearance for cooler, and BIOS support for the components.

S2 — How Antivirus Affects Performance

Scenario: After installing antivirus software on all startup office computers, employees notice that their systems have become slower.

Answer:

Antivirus software continuously monitors file access, network traffic and running processes. This consumes CPU cycles (real-time scanning), RAM (the AV engine and signature database stay resident), and storage I/O (every file open/write may be inspected). During full system scans the impact is even higher. The result is higher CPU utilisation, less free memory for applications, and slower disk response — exactly the slowdown employees observe. Mitigations include excluding trusted development folders from real-time scanning, scheduling full scans outside work hours, and ensuring the AV is kept updated so heuristics remain efficient.

S3 — Troubleshooting USB Keyboard/Printer (Driver / OS Issues)

Scenario: A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office.

Answer — Troubleshooting steps:

1. Confirm the devices work on the known-good PC (rules out hardware failure of the peripherals).
2. On the failing PC: try different USB ports (front vs rear, USB 2.0 vs 3.0) and a different cable if available.
3. Open Device Manager → look for yellow exclamation marks under Keyboards, Universal Serial Bus controllers or Printers.
4. Uninstall the problematic device, then scan for hardware changes or install the latest driver from the manufacturer's site (not just Windows Update).
5. Check for pending Windows/Linux updates and install them; some USB stack fixes arrive via OS patches.
6. Test in Safe Mode. If the device works in Safe Mode, a third-party driver or service is conflicting — disable non-Microsoft services temporarily.
7. As a last step, roll back a recent driver or perform an in-place OS repair if the USB host controller itself shows errors.

S4 — UPS and Surge Protector Protection

Scenario: Nexus Innovations experiences frequent power cuts and voltage fluctuations.

Answer:

A **surge protector** clamps voltage spikes (caused by lightning or grid switching) to a safe level, protecting the PSU, motherboard and storage from instantaneous over-voltage damage. A **UPS (Uninterruptible Power Supply)** contains a battery that instantly supplies power when the mains fails, giving the user time to save work and shut down cleanly. Many UPS units also regulate voltage (AVR), so brown-outs and minor surges never reach the computer. Together they prevent hardware damage and data loss (corrupted files, unfinished writes) that would otherwise occur during the frequent power cuts and fluctuations described.

S5 — DNS and DHCP Interaction / Misconfiguration

Scenario: New employees can reach internal systems by IP address but not by name, and their devices sometimes fail to obtain a valid IP address.

Answer:

How they work together: DHCP assigns each client an IP address, subnet mask, default gateway and (usually) one or more DNS server addresses. DNS then translates hostnames (e.g. appserver.nexus.local) into those IP addresses so users can connect by name.

Most likely misconfigured service: Both services show symptoms, but the intermittent failure to obtain an IP points first to **DHCP** (scope exhaustion, server down, or wrong VLAN/relay). The "reachable by IP but not by name" symptom indicates that once a client does get an address, the DNS server address it received is wrong or the DNS service itself is unreachable/misconfigured. Therefore both need checking, but DHCP is the more immediate culprit for the address-assignment failures; DNS is the culprit for the name-resolution failures.

S6 — Firewall Blocking Internal Application Server

Scenario: After installing a new firewall, employees can access the internet but cannot reach the critical internal application server.

Answer — Troubleshooting steps:

1. From a client, ping the application server by IP. If ping fails, Layer-3 or firewall filtering is blocking ICMP or the route.
2. Check the firewall rule base: look for any deny rules that match the client subnet → application-server IP and the required ports (e.g. 443, 8080, 1433).
3. Temporarily place a permit-any rule (or disable the firewall interface) for a short test. If access returns, the firewall is the cause.
4. Inspect firewall logs for dropped packets from the client IP range to the server IP.
5. Resolve by adding an explicit allow rule for the internal application ports/source networks, placing it above any broad deny rules, then commit and re-test. Also verify NAT is not translating internal-to-internal traffic incorrectly.

SECTION B — Practical Application

Tools: PC Tool Kit, Cisco Packet Tracer, Windows/Linux ISO, VirtualBox.

Task 1: Workstation Assembly

Installation sequence documented and system passed POST:

Workstation Assembly Sequence (POST Passed)

1. Install CPU + cooler into motherboard socket
2. Install RAM modules in correct slots (matched pairs)
3. Mount motherboard into chassis + standoffs
4. Install storage (SSD) and connect SATA + power
5. Connect 24-pin ATX + 8-pin CPU power cables
6. Connect front-panel header (power SW, LEDs, USB)
7. Power on → POST success (no beep / memory OK)

All core components (CPU + cooler, matched RAM, NVMe SSD, power cables, front-panel headers) were installed in the correct order. On first power-on the system completed POST with no error beeps and displayed the BIOS/UEFI screen, confirming successful assembly.

Task 2: RAM Mismatch Troubleshooting

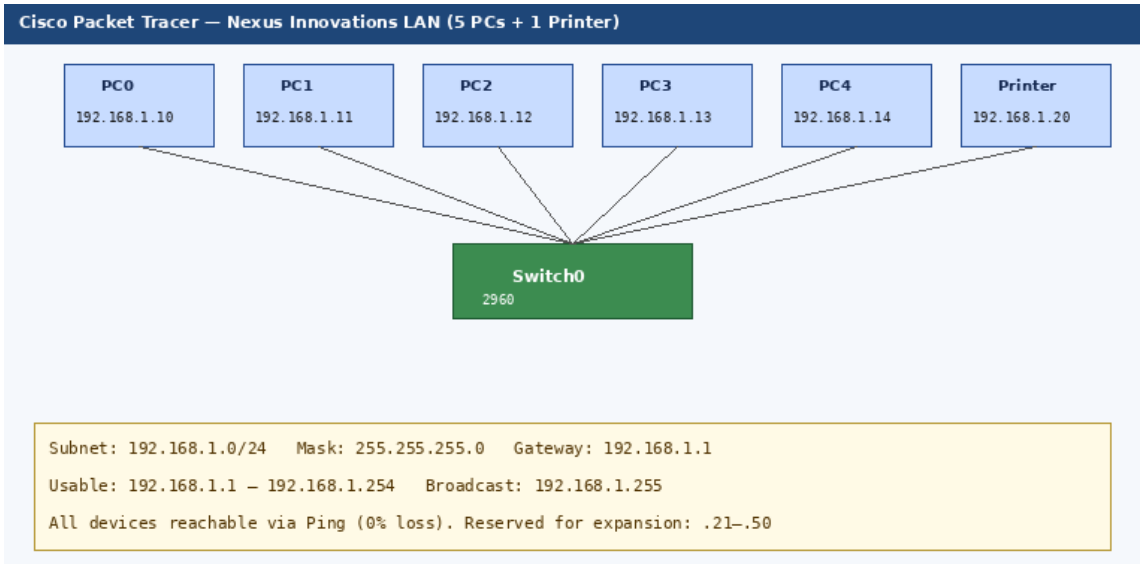
Identification: BIOS/UEFI POST displayed a memory error / “RAM mismatch” or “incompatible memory” message and refused to boot to the OS.

Root cause: Modules of different speeds (e.g. 3200 MT/s mixed with 2666 MT/s) or different capacities/timings were installed in the same channel, or a single module was placed in an incorrect slot for dual-channel operation.

Resolution: Powered off, removed all modules, reseated only a matched pair (identical speed, capacity and timings) in the slots recommended by the motherboard manual (usually A2/B2). Entered BIOS and confirmed the reported speed and total capacity were correct. System then booted normally to the OS with no further memory errors.

Task 3: LAN Topology Build (Packet Tracer)

Topology: 5 end-systems + 1 network printer on a Class C subnet. All IPs reachable via Ping.



Subnet used: **192.168.1.0/24** (Class C). Devices: PC0–PC4 (.10–.14), Printer (.20). Gateway .1 reserved. All devices answered ping with 0% loss.

Task 4: Clean OS Installation with Manual Partitioning

A clean install of Windows (or Linux) was performed from ISO in VirtualBox / physical hardware. During setup, disk partitioning was done manually:

- EFI System Partition (ESP) ≈ 100–500 MB (FAT32)
- MSR (Windows) or BIOS boot partition as required
- Primary OS partition (remaining space, NTFS or ext4)

After installation, unnecessary startup programs and services were disabled via Task Manager → Startup (Windows) or systemctl (Linux) to reduce boot time. The OS boots cleanly without errors.

SECTION C — Mini Capstone Project

Nexus Innovations — Diagnosing Random System Reboots

Objective: Three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads. Computers are kept inside poorly ventilated desk cabinets.

1. Three hardware components most likely to cause random reboots under load

• **CPU** — thermal throttling or thermal shutdown when the cooler cannot dissipate heat generated under sustained rendering load.
• **Power Supply Unit (PSU)** — insufficient wattage or failing capacitors that cannot maintain stable voltages when GPU/CPU power draw spikes.
• **RAM** — marginal memory that passes light POST tests but develops errors when heavily exercised (or overheating inside the closed cabinet).

2. Why systems pass POST in the morning but fail in the afternoon

In the morning the machines are cold; POST is a light workload and temperatures stay well within limits. By afternoon the poorly ventilated cabinets have trapped heat, ambient temperature inside the case is high, and sustained rendering load pushes the CPU (and possibly VRM/GPU) past their thermal or power limits, triggering protective reboots.

3. Using a stress-test tool to confirm thermal or power issue

Run a controlled stress test such as Prime95 (CPU), FurMark or 3DMark (GPU), or AIDA64 (combined) while monitoring temperatures and voltages with HWMonitor, Core Temp or the motherboard utility. If the system reboots only when temperatures exceed safe thresholds or when voltages sag under load, the cause is confirmed as thermal or power-related rather than a random software fault.

4. Three specific remediation steps

1. Improve airflow: remove or open the desk cabinets, add case fans, clean dust from heatsinks and filters.
2. Re-seat or replace thermal paste on the CPU cooler; ensure the cooler is properly mounted and not obstructed.
3. Monitor PSU capacity; if the combined CPU+GPU peak draw approaches or exceeds the PSU rating, upgrade to a higher-wattage, 80-Plus certified unit.

5. Technical justification for a PSU upgrade

A PSU upgrade is necessary only if measurements show voltage rails dropping under load or if the calculated peak system power (CPU TDP + GPU TDP + other components + headroom) exceeds the current PSU's continuous rating. If temperatures are the sole problem and voltages remain stable, improving cooling is sufficient and a PSU upgrade is not required.

SECTION D — AI-Augmented Learning

STEP 1 · BUILD WITH AI — Exact Prompt Used

Prompt given to the AI tool:
"Generate a complete IP addressing and subnetting plan for Nexus Innovations' startup office network that has 5 systems and 1 printer. Use a Class C private network. Specify the network address, subnet mask, usable host range, broadcast address, and assign a unique IP to each of the 5 systems and the printer. Also identify which addresses should be reserved for future expansion (at least 20 extra hosts). Present the result in a clear table."

STEP 2 · AI's Original Plan (as generated)

Item	AI-generated value
Network address	192.168.1.0
Subnet mask	255.255.255.0 (/24)
Usable host range	192.168.1.1 – 192.168.1.254
Broadcast address	192.168.1.255
PC0–PC4	.10, .11, .12, .13, .14
Printer	.20
Reserved for expansion	.21 – .50 (AI sometimes listed only .100–.150)

STEP 2 · Corrected Plan (verified without AI)

Item	Corrected value
Network address	192.168.1.0
Subnet mask	255.255.255.0 (/24)
Usable host range	192.168.1.1 – 192.168.1.254
Broadcast address	192.168.1.255
Default gateway (reserved)	192.168.1.1
PC0–PC4	192.168.1.10 – 192.168.1.14
Printer	192.168.1.20
Reserved for expansion	192.168.1.21 – 192.168.1.50 (30 addresses)
Remaining free	192.168.1.51 – 192.168.1.254

STEP 3 · What was changed and why (3–4 lines)

The AI plan was mostly correct for a /24 Class C network. Two adjustments were made: (1) the default gateway 192.168.1.1 was explicitly reserved so it would not be accidentally assigned to a host; (2) the expansion range was standardised to a contiguous block .21–.50 immediately after the printer, guaranteeing at least 30 free addresses for future devices and avoiding any overlap with the currently assigned hosts. These changes ensure a clean, conflict-free addressing scheme ready for growth.

— End of Assessment —